

On-Device Lava API (Android)

Status of this document (2026-06-02): Phases A, B, C, and D-infra of the *Lava API Android app* sub-project (spec [docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md](#), plan [docs/superpowers/plans/2026-06-02-lava-api-android-app.md](#)) have landed. This document describes **only what the committed code does** — the additive SQLite storage backend (Phase A), the in-process Go embed + c-shared/JNI native build (Phase B), the `:core:apiengine` Kotlin JNI wrapper (Phase C, commit `ef109760`), and the `:api-app` headless infrastructure (Phase D-infra, commit `e62e0fa8`: the foreground `ApiEngineService`, the `ApiEngineController` state machine, the `NsdManager` advertiser, and the per-install auth-key store). The **landing UI / control screen + ViewModel** (Phase D-ui), the instrumented **Compose UI Challenge tests** (Phase E), and the **client-side distinct labelling** of Android instances (sub-project 2) are **PENDING** and are explicitly marked as such below. Nothing in this doc describes behavior that is not yet in the tree.

Classification: project-specific (the on-device-API app + Go SQLite backend are Lava-specific; the additive-backend-selector pattern is universal and is noted as such inline).

1. Motivation

The existing Lava API (`lava-api-go`) runs on a host (server, NAS, or developer workstation) backed by Postgres, advertises itself on the LAN via mDNS, and the Android client discovers and talks to it over HTTPS. The on-device API sub-project lets a **phone or tablet itself become a LAN-reachable Lava API**: the same Go server, cross-compiled and embedded in-process, backed by a local SQLite file, bound to `0.0.0.0` so other devices on the same Wi-Fi can discover and use it.

The design constraint the operator set is *additive*: "Nothing already working can be broken — only extended for more flexibility." Every existing Postgres deployment behaves byte-for-byte as before; SQLite and the embed are new, config-selected branches.

2. The additive SQLite storage backend (Phase A — landed)

2.1 Principle (universal pattern)

A **backend selector** whose default reproduces today's behavior exactly. The existing code becomes "the Postgres branch," never "the only path." Any deployment that sets `LAVA_API_PG_URL` and nothing else behaves identically to before.

Classification: the selector pattern itself is universal; the two concrete backends are Lava-specific.

2.2 Config selection

Source: [lava-api-go/internal/config/config.go](https://github.com/lava-api-go/internal/config/config.go).

Variable	Meaning	Required when
<code>LAVA_API_STORAGE_BACKEND</code>	postgres (default) or sqlite	never (defaults to postgres)
<code>LAVA_API_PG_URL</code>	Postgres connection URL	iff backend is postgres
<code>LAVA_API_SQLITE_PATH</code>	SQLite database file path (e.g. <code>/data/lava-api.db</code> ; <code>:memory:</code> accepted)	iff backend is sqlite

`config.Load()` selects the branch in a switch `cfg.StorageBackend`:

- postgres → require `PGUrl` (the historical hard dependency, now scoped to this branch so existing users are unaffected).
- sqlite → require `SQLitePath`.
- anything else → reject with unknown storage backend %q.

No env var was renamed or removed; the two new variables carry placeholders in `.env.example` (§6.R). `LAVA_API_TLS_CERT/LAVA_API_TLS_KEY` and a valid `LAVA_API_MDNS_PORT` remain required regardless of backend.

2.3 Storage boundary and parity with Postgres

Source: [lava-api-go/internal/storage/](https://github.com/lava-api-go/internal/storage/).

The `storage.Storage` interface ([storage.go](#)) declares **exactly** the cache operations the handlers depend on — enumerated from the real handler consumers, nothing more:

```
type Storage interface {
    Get(ctx context.Context, key string) ([]byte, cache.Outcome,
        error)
    Set(ctx context.Context, key string, value []byte, ttl
        time.Duration) error
    Invalidate(ctx context.Context, key string) error
    Close() error
}
```

Two implementations satisfy it interchangeably:

- **postgres** — wraps the existing submodules/cache/pkg/postgres adapter (response_cache table, schema lava_api, 10-minute GC interval). Behavior is unchanged from the prior main.go path.
- **sqlite** — pure-Go [modernc.org/sqlite](#) (CGo-free, so it cross-compiles cleanly for Android). Behavior parity with Postgres is the explicit goal and is verified by a shared conformance harness plus a cross-backend parity test (per the plan).

`storage.New(cfg)` ([factory.go](#)) selects the implementation by `cfg.StorageBackend` and returns the `Storage` together with a `ReadinessFunc` (the /ready probe): Postgres uses `pgClient.HealthCheck`; SQLite does a trivial round-trip `Get` of a sentinel key to prove the handle answers queries.

2.4 SQLite TTL, GC, and WAL semantics

Source: [sqlite.go](#), migration [migrations/sqlite/0001_init.sql](#).

Schema mirrors the Postgres response_cache table with portable types:

```
CREATE TABLE IF NOT EXISTS response_cache (
    cache_key    TEXT PRIMARY KEY,
    value        BLOB NOT NULL,          -- BYTEA equivalent
    expires_at   INTEGER                 -- unix NANSECONDS; NULL =
    never expires
);
CREATE INDEX IF NOT EXISTS response_cache_expires_at_idx ON
    response_cache (expires_at);
```

Behavioral details the committed code implements:

- **TTL** is an absolute `expires_at` (unix nanoseconds). Set with `ttl > 0` stores `now + ttl`; `ttl <= 0` stores SQL NULL (never expires) — matching the Postgres `WHERE expires_at IS NULL OR expires_at > NOW()` contract.

- **Lazy expiry on read.** Get filters expires_at IS NULL OR expires_at > ?, so an expired entry reads as an OutcomeMiss, never a stale hit. A real DB error returns OutcomeBypass + err so callers fall through to the upstream — identical to cache.Client.Get. A stored zero-length value is a **hit** (non-nil empty slice), matching Postgres BYTEA empty-value semantics.
- **Background GC.** A goroutine runs sweepExpired every **10 minutes** (sqliteGCInterval, mirroring the Postgres GC cadence). It physically DELETes rows whose expires_at is in the past and then runs PRAGMA incremental_vacuum to return freed pages to the free list. Close() cancels the goroutine's context and joins it (no leak); Close() is idempotent.
- **WAL + auto_vacuum** (on-disk databases only; skipped for :memory:):
 - The DSN sets journal_mode(WAL) (concurrent readers don't block the single writer) and busy_timeout(5000) (ride out brief write locks).
 - SetMaxOpenConns(1) serializes access through one connection — the standard race-free choice for an embedded SQLite writer, and required so the connection-scoped auto_vacuum PRAGMA applies to the one connection every query uses.
 - PRAGMA auto_vacuum=INCREMENTAL followed by VACUUM is run **before** the migration (the VACUUM rewrites the DB header with the new flag; order is load-bearing — running the PRAGMA after a table exists leaves auto_vacuum=0).
 - After migration, PRAGMA journal_mode is re-queried and the constructor **fails loudly** if WAL did not engage, refusing to run in a weaker (DELETE) journal mode that would drop the concurrent-reader guarantee.

3. The in-process Go embed (Phase B — landed)

Source: [lava-api-go/internal/mobile/mobile.go](#) and [tls.go](#).

internal/mobile is the small, JNI-friendly lifecycle surface that boots the **full production API in-process** inside the Android app (Phases C/D will call it). Three exported functions, designed so only string/error cross the boundary:

```
func Start(configJSON string) error // parse config → SQLite +
    full router + TLS, listen, return when accepting
func Stop() error // graceful shutdown + close
    storage; errors if nothing running
func Status() string // JSON: state, scheme,
    bindAddr, port, requestCount, backend, version, auth*
```

Start's configJSON shape:

```
{"bindAddr":"0.0.0.0","port":8443,"sqlitePath":"/data/
x.db","authSharedKey":"<base64-uuid>","authFieldName":"<header>"}
— bindAddr/port optional (default 0.0.0.0:8443); sqlitePath required;
authSharedKey/authFieldName optional (see §3.4). Exactly one server
instance may run per process (a package-level handle guarded by a mutex;
Start while running returns an error).
```

3.1 Why c-shared, not gomobile

Source: [lava-api-go/cmd/lavaapi-cshared/main.go](#).

gomobile bind is **BLOCKED** on this module. lava-api-go's go.mod uses relative replace `../submodules/*` directives for the 16 vasic-digital submodules; gomobile's overlay-module generator cannot resolve them (it writes a 0-byte go.mod; proven on NDK 21.4 + 25.1). The chosen replacement is **go build -buildmode=c-shared**, which uses ordinary Go module resolution and therefore HONORS the replace directives. It is still an in-process, cross-compiled Go artifact.

The tradeoff: c-shared exports a flat C ABI, not a generated Java class hierarchy. A **hand-written JNI bridge** adapts the four exported C functions to the JNI naming convention the Kotlin side will bind.

The exported C surface (all strings NUL-terminated UTF-8; every returned `char*` is heap-allocated by `C.String` and the caller **MUST** release it via `LavaApiFree`):

```
extern char* LavaApiStart(char* configJSON); // "" on success,
      else error msg
extern char* LavaApiStop(void);              // "" on success,
      else error msg
extern char* LavaApiStatus(void);            // status JSON
      document
extern void  LavaApiFree(char* p);           // free a returned
      string
```

3.2 The JNI contract

Source: [lava-api-go/cmd/lavaapi-cshared/jni/jni_bridge.c](#) and [CMakeLists.txt](#).

`jni_bridge.c` builds `liblavaapi_jni.so`, which links against the prebuilt `liblavaapi.so` (from the c-shared build) and exposes the entry points the Kotlin side will bind:

```
package digital.vasic.lava.apigo
```

```
object LavaNative {
```

```

external fun nativeStart(configJson: String):
    String // "" on success, else error message
external fun nativeStop(): String // ""
    on success, else error message
external fun nativeStatus(): String //
    status JSON document
}

```

JNI symbol names (package dots → underscores, then class, then method):

- Java_digital_vasic_lava_apigo_LavaNative_nativeStart
- Java_digital_vasic_lava_apigo_LavaNative_nativeStop
- Java_digital_vasic_lava_apigo_LavaNative_nativeStatus

A Kotlin object compiles to a final class with a static INSTANCE; its external functions register as native methods that receive a jclass, so the bridge functions take (JNIEnv*, jclass, ...). The bridge copies each Go-returned C string into a JVM string via NewStringUTF and immediately releases the C buffer via LavaApiFree so nothing leaks across the boundary.

jni_bridge.c and its CMakeLists.txt live in the jni/ **subdirectory**, NOT the cgo main package, because jni_bridge.c #includes liblavaapi.h — the cgo-generated header the c-shared build PRODUCES — so it cannot be compiled by the cgo build itself; it is compiled later by the Android NDK toolchain via externalNativeBuild { cmake { ... } }.

LANDED (Phase C, commit ef109760): the Kotlin LavaNative object, the :core:apiengine wrapper module, and the externalNativeBuild Gradle wiring that loads these native libraries now exist in the tree and match the contract above exactly. See [§4A — Phase C](#) for the full mapping.

3.3 What the embed serves (full production router) + TLS

The embed builds the **EXACT same Gin engine** the production cmd/lava-api-go binary builds, via the shared internal/router.Build constructor ([router.go](#)) — search / browse / topic / forum / torrent / login / favorites / captcha, /v1/{provider}/..., plus /health and /ready. The provider registry wires the same adapters production wires (rutracker, nnmclub, kinozal, archive.org, gutenbergl). Sharing one router constructor is deliberate (§6.I): a divergent embed router would be a bluff vector — green tests against an embed serving different routes than production would guarantee nothing about production.

The **only** transport difference from the LAN binary: the embed serves HTTP/1.1 + HTTP/2 over TLS using a standard net/http server; the binary additionally serves HTTP/3 (QUIC). HTTP/3 is not required for the embed — LAN clients that discover the on-device API speak ordinary HTTPS. The

embed explicitly disables Brotli / protocol-metric / AltSvc middlewares (transport tuning, not security) so repeated in-process Start/Stop cycles don't touch the Prometheus default registerer.

TLS (tls.go):

- A self-signed **ECDSA P-256** cert + key is generated on first boot and persisted next to the SQLite DB (`lava-embed-cert.pem` / `lava-embed-key.pem`; the key is written `0600`). Validity is 10 years (rotation handling is a later sub-project).
- If both files already exist and parse, they are **reused** across restarts, so a peer that pinned the leaf cert out-of-band does not see it change every boot.
- **IP SANs**: loopback (IPv4 + IPv6) plus the host's non-loopback, non-link-local LAN IPs discovered at boot (`localIPs()`), so a peer addressing the server by LAN IP does not hit a host-mismatch error.
- **No wildcard DNS SAN**. DNS SANs are limited to `localhost` — a bare `*` wildcard SAN matched ANY hostname and defeated name verification (a security-review finding; fixed in commit `6f751495`). Peers address the embed by LAN IP (covered by the IP SANs) or by `localhost` on-device. Out-of-band leaf trust on the consuming devices is the model for now (full client-trust / local-CA handling is a later sub-project).

3.4 The Lava-Auth gate on the embed (same mechanism as the host)

Because the embed is network-exposed (§3.5), it **MUST authenticate** — there is no "trusted LAN" shortcut. It enforces the **identical** Lava-Auth gate the production LAN binary uses (`internal/auth.NewMiddleware + NewBackoffMiddleware`, source middleware.go):

- The client presents a credential in a configurable header (default Lava-Auth). The credential is `base64(16-byte UUID blob)`.
- The middleware `base64-decodes` the header, computes `hex(HMAC-SHA256(uuid_bytes, secret))`, and compares (in constant time, to defeat timing side-channels) against the active-clients allowlist.
- Active hash → request served (`client_name` set, backoff reset). Retired hash → 426 Upgrade Required. Missing / malformed / unknown → 401 and the per-IP backoff ladder advances; repeated failures escalate to 429 on the default ladder `2s, 5s, 10s, 30s, 1m, 1h`.
- `/health` and `/ready` are registered **before** the auth middleware so the orchestrator probes work without a credential.

The embed's app-managed key flow:

- The host app supplies the credential via `authSharedKey` in the Start config.

- **If none is supplied, the embed GENERATES one** (random 16-byte UUID, base64-encoded) so the gate is never disabled. The generated/in-effect key is surfaced via `Status()` (the `authKey` field) — and **only** there — so the app can display it for pairing. It is deliberately never written to a log line (§6.H / §6.AC redaction). The plaintext blob is zeroized after hashing.
- `Status().authEnabled` is always `true` while running.

LANDED (Phase D-infra, commit e62e0fa8): persisting the key across app restarts is now the `:api-app ApiKeyStore`'s responsibility (a per-install base64-UUID blob held in `EncryptedSharedPreferences`); see §4B.4 — auth-key store. The UI that shows it for pairing remains **PENDING (Phase D-ui)**. The Go embed surface here only **ENFORCES** whatever key is in effect and reports it via `Status()`.

3.5 Network exposure (0.0.0.0) and the trust boundary

The embed binds `0.0.0.0` by default — network exposure is the intended design. Loopback-only would defeat the feature (the whole point is that a phone running the embed becomes a LAN-reachable Lava API for tablets, TVs, and other phones). The bind address is validated with `net.ParseIP`; non-loopback (including the `0.0.0.0` wildcard) is explicitly allowed, and only unparseable IPs are rejected. The automated security-review suggestion of loopback-only was therefore explicitly overridden in the code, with the Lava-Auth gate (§3.4) as the compensating control: anyone on the LAN can *reach* the port, but only a holder of the auth key gets a non-401 response.

3.6 mDNS identity (Android instance vs host instance)

The embed is honestly the Go engine on Android with SQLite, so it will advertise:

- Service type `_lava-api._tcp` (release) / `_lava-api-dev._tcp` (debug) — the same types the client already watches (catalog in `core/data/.../DiscoveryServiceTypes.kt`: `SERVICE_TYPE_GO = "_lava-api._tcp"`, `SERVICE_TYPE_GO_DEV = "_lava-api-dev._tcp"`, `SERVICE_TYPE_KTOR = "_lava._tcp"`).
- TXT: `engine=go`, **`platform=android`**, `storage=sqlite`, `version=<n>`.

The existing host instance keeps advertising `engine=go` with `platform=server` (or absent) — unchanged; both appear in the client's list. The "distinct identity" is realized via the `platform` TXT key, which the client (sub-project 2) will label distinctly without any discovery re-architecture. Instances without `platform` render exactly as today (backward compatible).

LANDED (Phase D-infra, commit e62e0fa8): the on-device *NsdManager registration* (the embed advertising itself) now exists in `:api-app` as `NsdMdnsAdvertiser`; see §4B.3 — [mDNS advertiser](#). The client-side *discovery* that consumes these TXT records already exists in `LocalNetworkDiscoveryServiceImpl` and is described in `LOCAL_NETWORK_DISCOVERY.md`.

4A. Phase C — the `:core:apiengine` JNI wrapper (landed)

Source: `core/apiengine/` (commit ef109760).

`:core:apiengine` is an Android library that exposes the embedded `lava-api-go` server (the Go c-shared library + JNI bridge from Phase B) behind an ordinary Kotlin suspend-function API. Consumers (`:api-app`, future wiring) depend ONLY on this module's public surface — never on `LavaNative` directly. Per the Decoupled Reusable Architecture rule this is Lava-domain glue: it introduces no logic another vasic-digital project would consume.

4A.1 The public Kotlin surface (`ApiEngine`)

Source: `ApiEngine.kt` (package `lava.apiengine`).

```
interface ApiEngine {
    suspend fun start(config: ApiConfig): Result<ApiStatus> //
        // success → running ApiStatus; failure → native error
    suspend fun stop(): Result<Unit> //
        // failure when nothing is running (not a silent no-op)
    fun status(): ApiStatus //
        // non-suspending snapshot
}
```

`ApiConfig` is the start-time configuration, serialized to the exact JSON shape `internal/mobile.startConfig` parses:

Field	Type	Default	Meaning
<code>bindAddr</code>	<code>String</code>	<code>"0.0.0.0"</code>	bind IP; wildcard so LAN peers can reach the API (loopback-only would defeat the feature)
<code>port</code>	<code>Int</code>	8443	TCP listener port, matching the production LAN listener
<code>sqlitePath</code>	<code>String</code>	<i>(required)</i>	absolute path to the SQLite DB file; TLS material is persisted beside it
<code>authSharedKey</code>	<code>String?</code>	<code>null</code>	

Field	Type	Default	Meaning
			the base64-UUID credential; when null the embed GENERATES one and surfaces it via <code>ApiStatus.authKey</code> (there is no unauthenticated mode)
<code>authFieldName</code>	<code>String</code>	<code>"Lava-Auth"</code>	HTTP header the credential is read from

`ApiStatus` mirrors the JSON `internal/mobile.Status()` returns: `state` ("running"/"stopped"), `bindAddr`, `port`, `requestCount`, `backend` (always "sqlite"), `version`, `scheme` (always "https"), `authEnabled` (always true while running), `authFieldName`, and the nullable `authKey` (the in-effect credential the host app can display for pairing; null when stopped).

Exactly one server instance runs per process (enforced by the native side): start while already running returns a failed `Result`.

4A.2 NativeApiEngine — the real JNI-backed implementation

Source: [NativeApiEngine.kt](#).

`NativeApiEngine` is the production `ApiEngine`. It maps the Kotlin surface onto the three `LavaNative` external functions:

- `start(config)` serializes `config` to JSON via `kotlinx.serialization` (`ConfigDto` — field names `bindAddr`/`port`/`sqlitePath`/`authSharedKey`/`authFieldName` matching the Go `startConfig` tags; a null `authSharedKey` is emitted as "", which the Go side treats as "generate one"), calls `LavaNative.nativeStart(configJson)`, and — because the native `Start/Stop` contract is "empty string on success, error message otherwise" — maps a non-empty return to `Result.failure(ApiEngineException(err))`. On success it parses `LavaNative.nativeStatus()` into `ApiStatus`.
- `stop()` calls `LavaNative.nativeStop()` and applies the same empty-string = success mapping.
- `status()` parses `LavaNative.nativeStatus()` (the `StatusDto` fields are empty on the Go side when stopped, so they default to stopped-state zero values).

`start` and `stop` run on an injectable `CoroutineDispatcher` (default `Dispatchers.IO`) because the native calls may block on socket bind / TLS material generation / SQLite open. The class requires a real Android device/emulator (the native .so is Android-only), so it is NOT JVM-unit-testable;

its behavioral contract is verified by the Phase E Compose UI Challenge against a real emulator (PENDING). The JVM-testable contract lives in FakeApiEngine.

4A.3 LavaNative — the JNI binding

Source: `LavaNative.kt` (package `digital.vasic.lava.apigo` — chosen to match the JNI symbol encoding).

package digital.vasic.lava.apigo

```
internal object LavaNative {  
    init { System.loadLibrary("lavaapi_jni") }  
    external fun nativeStart(configJson: String):  
        String // "" on success, else error message  
    external fun nativeStop(): String // ""  
        on success, else error message  
    external fun nativeStatus(): String //  
        status JSON document  
}
```

The package + object + method names MUST match the JNI symbol encoding in `jni_bridge.c` exactly (e.g. `Java_digital_vasic_lava_apigo_LavaNative_nativeStart`). `System.loadLibrary("lavaapi_jni")` resolves to `liblavaapi_jni.so`; its transitive dependency on `liblavaapi.so` (the prebuilt Go c-shared library) is satisfied because BOTH libraries are packaged into the same `jniLibs/<abi>/` directory of the APK. The object is internal — consumers go through `NativeApiEngine`, never here.

4A.4 FakeApiEngine — the behaviorally-equivalent JVM fake

Source: `FakeApiEngine.kt`.

Per the Anti-Bluff Pact Third Law, the fake reproduces the production / Go BRANCHES that affect callers — it is NOT a "simpler than reality" stub:

- **Single instance.** start while already running fails with an "already running" error matching the Go "mobile: server already running ...".
- **Stop is not a no-op.** stop without a running server fails with a "no server running" error matching `internal/mobile.Stop`.
- **Status reflects last start/stop.** stopped before any start; running with the started config's `bindAddr/port` after start; reverts to stopped after stop.
- **Auth always on while running.** start surfaces an auth key (the config's `authSharedKey` when provided, else a deterministic generated stand-in) and reports `authEnabled == true` — no unauthenticated mode.

- **Error injection.** `failWith(error)` makes the next start/stop propagate a `Result.failure`, exercising callers' error paths.
`recordRequests(count)` simulates inbound traffic so `status()` reports a non-zero count.

4A.5 The native build pipeline (`buildCshared` → `jniLibs` → `externalNativeBuild`)

Source: [core/apiengine/build.gradle.kts](#).

Two native libraries are packaged per ABI into this module's AAR, for the three ABIs `arm64-v8a` / `x86_64` / `armeabi-v7a` (NDK 25.1.8937393, CMake 3.22.1):

1. **`liblavaapi.so`** — the prebuilt Go c-shared library. A `buildCshared` Gradle Exec task shells out to `lava-api-go/scripts/build-cshared.sh` for the module's ABIs, writing `lava-api-go/build/jniLibs/<abi>/liblavaapi.{so,h}`. The task declares those `.so/.h` as outputs so it is **up-to-date when all expected outputs are already present** — a clean assemble does not force a multi-minute Go rebuild when the artifacts are fresh. On a Windows build host the task fails honestly (the Go c-shared/JNI path is unsupported there) rather than producing a broken artifact. `sourceSets.main.jniLibs.srcDir(prebuiltJniLibsDir)` points AGP at that output dir so it packages `liblavaapi.so` per ABI directly (Go is NOT recompiled from Gradle — the script output is LOCATED and staged).
2. **`liblavaapi_jni.so`** — the hand-written JNI bridge. AGP's `externalNativeBuild { cmake { path = jniCmakeLists } }` points at the canonical `lava-api-go/cmd/lavaapi-cshared/jni/CMakeLists.txt`, which imports `liblavaapi.so` as an IMPORTED SHARED lib and links the bridge against it. `-DLAVAAPI_PREBUILT_DIR=<abs path>` is passed as a CMake argument so the `CMakeLists` finds the prebuilt `.so/.h` (laid out as `<ABI>/liblavaapi.{so,h}`).

The `native-build / jniLibs-merge` tasks (`externalNativeBuild*`, `configureCMake*`, `buildCMake*`, `mergeJniLibFolders`) all `dependsOn(buildCshared)` so the prebuilt inputs exist before CMake configures and before the `jniLibs` merge reads the directory.

4B. Phase D-infra — the `:api-app` module (landed)

Source: [api-app/](#) (commit `e62e0fa8`).

:api-app is the standalone *Lava API* Android application. It hosts the embedded lava-api-go server (via :core:apiengine) as an on-device, LAN-reachable HTTPS API, advertised over mDNS so other devices discover it. Phase D-infra delivers the **headless infrastructure** only: the foreground Service, the Android-free lifecycle controller, the mDNS advertiser, and the per-install auth-key store. The Compose landing UI, the refined notification copy, and the ViewModel are **Phase D-ui (PENDING)**; this module ships a placeholder MainActivity (its onCreate calls setContent { }) so the manifest is valid and the APK assembles.

4B.1 Module identity, signing, and permissions

Source: [api-app/build.gradle.kts](#), [AndroidManifest.xml](#).

- **Application ID** digital.vasic.lava.api; the debug build applies applicationIdSuffix = ".dev" → digital.vasic.lava.api.dev, so the release and dev variants install side-by-side and advertise distinct mDNS service types (§4B.3). versionCode = 1, versionName = "0.1.0", minSdk = 23 (the per-install auth-key store relies on EncryptedSharedPreferences, which requires API 23+; the client :app keeps minSdk 21).
- **Shared signing.** The signing block mirrors :app's .env-driven inputs verbatim — the SAME KEYSTORE_PASSWORD + KEYSTORE_ROOT_DIR keystore material (debug → keystores/debug.keystore alias debug; release → keystores/release.keystore alias release). No parallel signing scheme is invented (§6.R / §6.H). Release uses postprocessing (remove-unused-code + remove-unused-resources + optimize, isObfuscate = false).
- **Permissions** the embed needs while serving: INTERNET, ACCESS_NETWORK_STATE, ACCESS_WIFI_STATE, CHANGE_WIFI_MULTICAST_STATE (mDNS multicast), FOREGROUND_SERVICE + FOREGROUND_SERVICE_DATA_SYNC, POST_NOTIFICATIONS, and WAKE_LOCK. The application sets android:usesCleartextTraffic="true" (matching the client :app). The Service is declared android:exported="false" with android:foregroundServiceType="dataSync" (the closest standard type for a process that continuously serves LAN requests + advertises over mDNS). The ApiApplication is a @HiltAndroidApp root; D-infra wires no Hilt modules (the Service constructs its controller directly) — the annotation is present so D-ui can add injected ViewModels without re-shaping the app.

4B.2 The foreground ApiEngineService + OS locks

Source: [service/ApiEngineService.kt](#).

A foreground Service that owns the Android-free ApiEngineController and drives it via intent actions:

- **Intent actions:** ACTION_START (lava.api.app.action.START, the default), ACTION_STOP, ACTION_RESTART. onStartCommand promotes to foreground *immediately* (startForeground with a baseline notification) so the OS does not kill the process before the first state emission, then dispatches the matching controller.start() / stop() / restart() on a service-scoped CoroutineScope (cancelled in onDestroy). It returns START_STICKY.
- **OS locks held while serving** (acquired on Running, released on Stopped / Error / onDestroy):
 - a Wi-Fi **MulticastLock** so mDNS multicast reaches the LAN;
 - a Wi-Fi **wifiLock** in WIFI_MODE_FULL_HIGH_PERF so the radio stays up;
 - a **partial WakeLock** (PARTIAL_WAKE_LOCK) so the CPU keeps serving while the screen is off. All three are setReferenceCounted(false) and released the moment the API stops (idempotent release guarded by isHeld).
- **The controller is built** with a NativeApiEngine, an NsdMdnsAdvertiser whose engine identity is GO_DEV when the package name ends in .dev else GO, an ApiKeyStoreImpl.create(applicationContext), a NetworkInterfaceLanIpProvider, and a sqlitePath of filesDir/lava-api.db.
- **Foreground notification:** an ongoing, low-importance notification on channel lava.api.app.server. When Running the body shows "<url>
• <n> req" (reachable URL + live request count); it carries **Stop** + **Restart** actions (each a PendingIntent.getService with the matching action) and a content-intent back to MainActivity. The copy is intentionally minimal (Phase D-ui refines it).

4B.2a The ApiEngineController state machine

Source: [control/ApiEngineController.kt](#), [control/ApiControlState.kt](#).

The controller is the **Android-free lifecycle core** — it has NO Service / Context dependency, so it is unit-tested with the real controller wired to FakeApiEngine + fakes for the advertiser/keyStore (the load-bearing Anti-Bluff test ApiEngineControllerTest). It owns the ApiEngine, the MdnsAdvertiser, and the ApiKeyStore, and exposes state: StateFlow<ApiControlState> as the single source of truth the Service + UI observe.

The ApiControlState sealed interface forms the cycle Stopped → Starting → Running → Stopping → Stopped, with Error reachable from Starting/Stopping:

State	Meaning
Stopped	no server running (initial state)
Starting	a start was requested; the engine has not yet reported Running
Running	the embed is up; carries url, lanIps, port, requestCount, authKey, authFieldName — all from the real post-start ApiStatus + the host's discovered LAN IPs, never synthesized
Stopping	a stop was requested; shutdown not yet confirmed
Error(message)	the last start/stop failed; message is the engine's error text

Behavioral details the committed code implements:

- **start()** emits Starting, fetches the persisted key via `keyStore.getOrCreate()`, builds an `ApiConfig` (port, `sqlitePath`, the key as `authSharedKey`, `keyStore.fieldName` as `authFieldName`), and calls `engine.start(config)`. On success it calls `onStarted(status)`: it resolves the host's LAN IPs (`lanIpProvider()`), **registers the mDNS advertisement on the running port**, and emits Running with the reachable URL built as `<scheme>://<first LAN IP or 127.0.0.1>:<port>`. On engine failure it emits Error and does **NOT** advertise.
- **stop()** emits Stopping, **unregisters the advertisement first** (so a stale record never points at a dead listener), then calls `engine.stop()`: success → Stopped, failure → Error.
- **restart()** is `stop()` then `start()`; if the stop surfaced an Error it halts there (a failed stop leaves the embed in an unknown state; surfacing Error is the honest behavior).
- The default port is read from `ApiConfig`'s own default so the literal lives in `:core:apiengine`, not duplicated here (§6.R). The host's LAN IPs come from `NetworkInterfaceLanIpProvider` ([net/LanIpProvider.kt](#)), which enumerates the real non-loopback, non-link-local IPv4 interface addresses — not a synthesized value (Sixth-Law clause 3: the URL the UI shows and the cert SANs the embed generates both depend on these being the TRUE reachable IPs).

4B.3 The mDNS advertiser (NsdMdnsAdvertiser)

Source: [service/MdnsAdvertiser.kt](#), [service/NsdMdnsAdvertiser.kt](#).

`MdnsAdvertiser` is a two-method interface (`register(port) / unregister()`); `NsdMdnsAdvertiser` is the `NsdManager`-backed implementation. It registers a single `NsdServiceInfo` named "Lava API" for the build-appropriate service type, on the running port, with the TXT

records the pure `buildTxtRecords` builder produces. `register` first calls `unregister()` (a previous registration is torn down before re-registering); `unregister` is safe to call when not registered.

- **Service types** (matching `DiscoveryServiceTypeCatalog` in `core/data` — the cross-process mDNS protocol contract):
 - release build → `_lava-api._tcp` (engine identity `G0`)
 - debug build → `_lava-api-dev._tcp` (engine identity `G0_DEV`)
- **TXT records** (the pure, unit-testable wire contract via `buildTxtRecords` — verified by `MdnsAdvertiserTxtTest` without touching Android):
engine=`go` (or `go-dev`), platform=`android`, storage=`sqlite`, version=`<n>`. The version is read from a `versionProvider` lambda (`engine.status().version`) at register time so the advertised version tracks the actual running embed, not a hardcoded literal (§6.R).

The service-type + TXT literals are kept in `:api-app` (rather than depending on `:core:data`) so the API app does not pull in the entire client data layer; they are identical to `DiscoveryServiceTypeCatalog.SERVICE_TYPE_G0 / SERVICE_TYPE_G0_DEV`.

4B.4 The per-install auth-key store (`ApiKeyStore`)

Source: [auth/ApiKeyStore.kt](#).

`ApiKeyStore` is a two-member interface — `getOrCreate(): String` and `fieldName: String`. `ApiKeyStoreImpl` implements it:

- The credential is the **production wire shape the Go embed accepts**: the base64 (standard alphabet, padded, single line) encoding of a random 16-byte UUID blob. This is EXACTLY what `internal/mobile.generateAuthKey` mints and what `internal/auth.NewMiddleware` base64-decodes before HMAC-ing — so a key produced here is honoured by the embed when passed as `ApiConfig.authSharedKey`.
- **First run** generates a fresh key with `SecureRandom` (16 bytes, `UUID_LEN`), base64-encodes it (`android.util.Base64.NO_WRAP`, matching Go's `base64.StdEncoding`), and persists it; every subsequent call (this run or a later install lifetime) returns the SAME key so paired peer devices keep working. `getOrCreate` is `@Synchronized`.
- **Encrypted at rest**: `ApiKeyStoreImpl.create(context)` builds an `EncryptedSharedPreferences` (AES256-SIV key encryption + AES256-GCM value encryption, AES256_GCM master key) in the `lava_api_auth` prefs file. Pure- JVM unit tests construct `ApiKeyStoreImpl` directly with an in-memory `SharedPreferences` fake + a `java.util.Base64`-backed `KeyEncoder` that yields the identical alphabet (the encoder is injectable for exactly this reason).

- **\$6.H credential hygiene:** the key is NEVER written to a log line, never concatenated into an exception message, never returned from toString. The plaintext blob is fill(0)'d in a finally block after encoding; the base64 String leaves the process only through the deliberate display/config path (D-ui). The default fieldName is "Lava-Auth", matching the embed.

PENDING (Phase D-ui): the landing screen / control screen, the ViewModel, and the refined notification copy that DISPLAY the URL, the auth key (for pairing), and the live request count are not yet in the tree — MainActivity is a placeholder. **PENDING (Phase E):** the instrumented Compose UI Challenge tests that boot the embed on a real emulator and issue a real HTTPS request. **PENDING (sub-project 2):** the client-side distinct labelling of platform=android instances in the discovery list.

4. Diagrams

4.1 Component / module diagram (what has landed)

```

flowchart TB
    subgraph LavaApiGo["lava-api-go (Go)"]
        cfg["internal/config<br/>storage selector<br/>postgres |
sqlite"]
        router["internal/router.Build<br/>FULL production Gin
engine<br/>(shared by binary + embed)"]
        auth["internal/auth<br/>Lava-Auth gate<br/>HMAC-SHA256 +
backoff ladder"]
        subgraph Storage["internal/storage"]
            iface["Storage interface<br/>Get / Set / Invalidate /
Close"]
            pg["postgres impl<br/>(submodules/cache)"]
            sq["sqlite impl<br/>modernc.org/sqlite<br/>WAL +
INCREMENTAL auto_vacuum + GC"]
        end
        mobile["internal/mobile<br/>Start / Stop / Status<br/>
>0.0.0.0 + self-signed TLS"]
        cshared["cmd/lavaapi-cshared<br/>C ABI: LavaApiStart/Stop/
Status/Free"]
    end

    subgraph NativeBuild["c-shared build (build-cshared.sh)"]
        so["liblavaapi.so + liblavaapi.h<br/>per ABI: arm64-v8a /
x86_64 / armeabi-v7a"]
        jni["jni_bridge.c + CMakeLists.txt<br/>liblavaapi_jni.so"]
    end

    subgraph Android["Android :api-app + :core:apiengine"]
        kotlin["LavaNative (external fun)<br/>(Phase C - landed)"]
    end

```

```

        engine["core:apiengine<br/>ApiEngine /
NativeApiEngine<br/>(Phase C - landed)"]
        ctrl["ApiEngineController<br/>Stopped/Starting/Running/
Stopping/Error<br/>(Phase D-infra - landed)"]
        svc["ApiEngineService (foreground)<br/>Wifi/Multicast/Wake
locks<br/>(Phase D-infra - landed)"]
        adv["NsdMdnsAdvertiser<br/>_lava-api(-dev)._tcp + TXT<br/
>(Phase D-infra - landed)"]
        keys["ApiKeyStore<br/>EncryptedSharedPreferences<br/
>(Phase D-infra - landed)"]
        ui["ApiControl landing UI + ViewModel<br/>(Phase D-ui -
PENDING)"]
    end

    cfg --> Storage
    iface --> pg
    iface --> sq
    mobile --> router
    router --> auth
    router --> iface
    mobile --> sq
    cshared --> mobile
    cshared --> so
    so -.includes.-> jni
    jni -- JNI --> kotlin
    kotlin --> engine
    svc --> ctrl
    ctrl --> engine
    ctrl --> adv
    ctrl --> keys
    svc -. --> ui

    classDef pending fill:#eee,stroke:#999,stroke-dasharray: 5 5;
    class ui pending;

```

4.2 Start / Stop / Restart lifecycle state machine

```

stateDiagram-v2
    [*] --> Stopped
    Stopped --> Starting: Start(configJSON)
    Starting --> Running: listener accepting<br/>(net.Listen
returned)
    Starting --> Error: bind / storage / TLS failure
    Running --> Stopping: Stop()
    Stopping --> Stopped: graceful shutdown + store.Close()
    Running --> Running: Start() while running → error<br/
>(exactly one instance per process)
    Error --> Starting: Start(configJSON) retry
    Stopped --> Stopped: Stop() → error "no server running"

    note right of Running
        Status() reports:
        state=running, scheme=https,

```

```

        bindAddr, port, requestCount,
        backend=sqlite, version,
        authEnabled=true, authFieldName, authKey
    end note

```

The Go embed itself exposes only Stopped and Running via `Status().state`. The Starting/Stopping/Error transient states are now realized in Kotlin by the `:api-app ApiEngineController` (Phase D-infra — landed; see §4B.2a): the controller emits Starting before `engine.start()` and Stopping before `engine.stop()`, and Error(message) when either fails. The UI that RENDERS these states is PENDING Phase D-ui.

4.3 mDNS discovery + authenticated request sequence

sequenceDiagram

participant P as Phone running :api-app
(ApiEngineService + controller — landed)

participant N as LAN (mDNS / multicast)

participant C as Other Lava client

P->>P: Service.start → controller.start → engine.start
>(0.0.0.0:8443, SQLite, self-signed TLS)

P->>P: ApiKeyStore.getOrCreate → Lava-Auth key (persisted, EncryptedSharedPreferences)

Note over P,N: Phase D-infra — NsdMdnsAdvertiser.register (NsdManager.registerService)

P->>N: advertise _lava-api._tcp
>TXT engine=go, platform=android, storage=sqlite, version=n

C->>N: NsdManager discover _lava-api._tcp (already in-tree)

N-->>C: resolved host:port + TXT

C->>C: map to Endpoint.GoApi(host, port)
>label platform=android distinctly (PENDING sub-project 2)

C->>P: HTTPS GET /health (no auth required)

P-->>C: 200 OK

C->>P: HTTPS GET /v1/{provider}/search?... (Lava-Auth: <key>)

P->>P: auth middleware: HMAC-SHA256 vs allowlist (constant-time)

alt valid key

P-->>C: 200 + real JSON

else missing / wrong key

P-->>C: 401; backoff ladder advances → 429 on repeat

end

4.4 Build pipeline (Go → per-ABI .so → jniLibs → app)

flowchart LR

src["internal/mobile +
cmd/lavaapi-cshared"]

subgraph build["build-cshared.sh (per ABI)"]

cc["CGO_ENABLED=1
GOOS=android GOARCH=<arch>
>CC=NDK clang"]

```

        gob["go build -buildmode=c-shared"]
        verify["llvm-nm -D → assert LavaApiStart present<br/>+
sha256 + size"]
    end
    out["build/jniLibs/&lt;abi&gt;/<br/>liblavaapi.{so,h}<br/>
(gitignored)"]
    task["core:apiengine buildCshared task<br/>(invokes build-
cshared.sh;<br/>up-to-date when outputs fresh)"]
    cmake["AGP externalNativeBuild → NDK CMake<br/>jni_bridge.c →
liblavaapi_jni.so<br/>(Phase C – landed)"]
    jnilibs["core:apiengine jniLibs srcDir +<br/>
externalNativeBuild output<br/>liblavaapi.so + liblavaapi_jni.so
per ABI<br/>(Phase C – landed)"]
    app[":api-app APK (depends on :core:apiengine)<br/>(Phase D-
infra – landed)"]

src --> cc --> gob --> verify --> out
task --> out
out --> cmake --> jnilibs --> app

```

5. Building the native libraries

See the script guide <docs/scripts/build-cshared.sh.md> for the authoritative reference. In brief:

```

cd lava-api-go
./scripts/build-cshared.sh                # all three ABIs
./scripts/build-cshared.sh arm64-v8a     # just the critical ABI

```

ABI	GOARCH	NDK clang wrapper (API=28 default)
arm64-v8a	arm64	aarch64-linux-android28-clang
x86_64	amd64	x86_64-linux-android28-clang
armeabi-v7a	arm	armv7a-linux-androideabi28-clang

Outputs land in `lava-api-go/build/jniLibs/<abi>/liblavaapi.{so,h}` (gitignored — only the source is committed). The script prints each `.so`'s size + sha256 and asserts `LavaApiStart` is in the dynamic symbol table via `llvm-nm`. A missing NDK toolchain or NDK clang is reported as a hard failure (exit 2 for config errors, 1 for build/symbol failures) — no `.so` is faked (§6.I / Anti-Bluff).

LANDED (Phases C + D-infra): the `:core:apiengine` wrapper (Phase C, commit `ef109760`) is now the Gradle consumer of these native libraries — its `buildCshared` task invokes `build-cshared.sh` and its `externalNativeBuild` packages `liblavaapi.so` + `liblavaapi_jni.so` per ABI (see §4A.5). The `:api-app` module (Phase D-infra, commit

e62e0fa8) depends on `:core:apiengine` and ships the foreground service + controller + on-device mDNS advertiser + auth-key store (see §4B).

PENDING (Phases D-ui / E): the landing UI + control screen + ViewModel (Phase D-ui) and the instrumented Compose UI Challenge tests that boot the embed on a real emulator and issue a real HTTPS request (Phase E) are not yet in the tree.

5A. API[?]embed sync guarantee (no drift) — §11.4.69 / §6.J

The on-device embed (`liblavaapi.so`) MUST contain EXACTLY the current `lava-api-go` source codebase — a stale `.so` would serve old API behaviour while every other gate looked green (the canonical "tests pass, wrong-version feature" bluff). A four-part mechanism, anchored on ONE hash function, makes drift mechanically impossible to hide:

1. **Single source of truth — `scripts/compute-api-source-hash.sh`.** A deterministic bare-64-hex sha256 over the EXACT file set that compiles into the `.so`: every non-test `.go` under `lava-api-go/cmd/lavaapi-cshared + lava-api-go/internal`, plus `go.mod + go.sum`. The list is byte-sorted and each file's relative path + content is hashed, so a rename, move, content edit, or dependency bump each change the digest; `*_test.go` and the other `cmd/` entrypoints are excluded (they do not link into the embed).
2. **Build-time injection + committed manifest — `build-cshared.sh`.** The build computes the hash via (1) and injects it into `internal/version.SourceHash` via `-ldflags -X`, so the RUNNING `.so` reports it through `mobile.Status()` (`sourceHash` field → Kotlin `ApiStatus.sourceHash`). On a successful build it also writes the committed manifest `core/apiengine/src/main/resources/api-source.hash`.
3. **CI gate — `scripts/check-api-app-sync.sh`.** Wired into `scripts/ci.sh` (both `--changed-only` and `--full`). Recomputes the live hash and compares it to the committed manifest; on mismatch it exits 1 loudly (on-device API embed is STALE vs `lava-api-go - rebuild liblavaapi.so` via `build-cshared.sh`). Pure bash + git, fast, no failure-swallow (§6.J).
4. **Gradle input-correctness — `core/apiengine/build.gradle.kts`.** The `buildCshared` task declares the `lava-api-go` embed-source dirs + `go.mod/go.sum` as inputs (and the manifest as an output), so Gradle re-runs `build-cshared.sh` on ANY embed-source change — killing the stale-

cache drift at the root. (Before this fix the task declared only the script as an input and went up-to-date whenever the .so outputs existed, even after the Go source changed.)

Runtime proof on device — api-app Challenge C05.

Challenge05ApiEmbedSourceHashMatchesTest starts the real embed (real UI → real controller → real NativeApiEngine → real .so) and asserts the running embed's ApiStatus.sourceHash equals the APK's BuildConfig.LAVA_API_SOURCE_HASH (both computed from the SAME script), with an empty-hash hard-fail. Equal hashes prove the packaged embed equals the source the APK was built against.

Go contract test. lava-api-go/tests/contract/sourcehash_contract_test.go asserts hash stability, manifest-matches-live, and falsifiability (a content edit to a tracked embed-linked .go changes the hash).

Per-script guides: [docs/scripts/compute-api-source-hash.sh.md](#) and [docs/scripts/check-api-app-sync.sh.md](#).

6. Constitutional alignment (what the landed code already satisfies)

- **\$6.R No-Hardcoding** — SQLite path, bind addr, port, header name, and NDK path/API level all come from config / env / Start-config, never literals.
- **\$6.H Credential Security** — the auth key is surfaced only via Status(), never logged; plaintext UUID blobs are zeroized after hashing.
- **\$6.J Anti-Bluff** — the embed reuses the production router so embed tests guarantee production behavior; the build script reports real .so evidence; FakeApiEngine reproduces the production branches (single-instance, stop-not-a-no-op, auth-always-on) rather than being a simpler stub; ApiEngineController is Android-free so its load-bearing test wires the REAL controller to the fake engine; Running state fields come from the real ApiStatus + real LAN IPs, never synthesized.
- **\$6.U No sudo/su** — the build uses user-local NDK tooling only.

What Phases C + D-infra add

- **\$6.R No-Hardcoding** — the controller's default port is read from ApiConfig's own default (the literal lives in :core:apiengine), the advertised version is read from engine.status().version, the LAN IPs are the real interface addresses, and :api-app's signing reuses :app's .env-driven keystore inputs rather than inventing keys.

- **§6.H Credential Security** — ApiKeyStore holds the credential in EncryptedSharedPreferences, zeroizes the plaintext UUID blob after encoding, and never logs / toStrings / exception-concatenates the key; it is surfaced only through the deliberate D-ui display path.

7. References

- Spec: [docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md](#)
- Plan: [docs/superpowers/plans/2026-06-02-lava-api-android-app.md](#)
- Script guide: [docs/scripts/build-cshared.sh.md](#)
- User guide: [docs/guides/ON_DEVICE_API_USER_GUIDE.md](#)
- Discovery flow: [docs/LOCAL_NETWORK_DISCOVERY.md](#)
- Architecture overview: [docs/ARCHITECTURE.md](#)